

INTEGRAL CALCULUS

Indefinite integrals

Integration is the inverse process of differentiation. This chapter is devoted to indefinite integrals which is the second major problem of calculus.

Antiderivatives

Definition: A function F is called an *antiderivative* of a function f on a given interval if $F'(x) = f(x)$ for all values of x in that interval. The functions $(1/3)x^3$, $(1/3)x^3 + \sqrt{5}$, $(1/3)x^3 + C$, etc. are antiderivatives of $f(x) = x^2$ on the interval $(-\infty, +\infty)$.

The process of computing antiderivatives is called *antidifferentiation* or *integration*. If $\frac{d}{dx}F(x) = f(x)$ then the functions of the form $F(x) + c$ are the antiderivatives of $f(x)$. It is denoted by $\int f(x)dx = F(x) + c$, where the symbol $\int$ is called the *integral sign* and $f(x)$ is the *integrand*. The constant c is called the *integrating constant*. Since the right side of this relation is not a definite function the term '*indefinite*' is used.

Properties

Let $F(x)$ and $G(x)$ be two antiderivatives of $f(x)$ and $g(x)$, respectively and let c be a constant, real or complex. Then the following properties hold:

- (i) $\int cf(x)dx = cF(x) + C$
- (ii) $\int [f(x) + g(x)]dx = F(x) + G(x) + C$
- (iii) $\int [f(x) - g(x)]dx = F(x) - G(x) + C$

Problems

For necessary formulas see page 324, Table 5.2.1 of the textbook Calculus by Howard Anton, Irl Bivens and Stephen Davis (10th Edition)

1. Evaluate $\int \frac{t^2 - 2t^4}{t^4} dt$

Solution: We can write $\int \frac{t^2 - 2t^4}{t^4} dt = \int \left(\frac{t^2}{t^4} - \frac{2t^4}{t^4} \right) dt = \int (t^{-2} - 2) dt = -\frac{1}{t} - 2t + C$

2. Evaluate (i) $\int \frac{1 - 2t^3}{t^3} dt$, (ii) $\int \left(\frac{1}{2t} - \sqrt{2}e^t \right) dt$

3. Evaluate $\int \frac{x^2}{x^2 + 1} dx$

Solution: We can write $\int \frac{(x^2 + 1) - 1}{x^2 + 1} dx = \int 1 dx - \int \frac{1}{x^2 + 1} dx = x - \tan^{-1} x + C$

4. Evaluate $\int [\operatorname{cosec}^2 t - \sec t \tan t] dt$

Solution: We can write $\int [\operatorname{cosec}^2 t - \sec t \tan t] dt = -\cot t - \sec t + C$

5. Evaluate $\int \operatorname{cosec} x (\sin x + \cot x) dx$

Solution: We can write $\int \operatorname{cosec} x (\sin x + \cot x) dx = \int (1 + \operatorname{cosec} x \cot x) dx = x - \operatorname{cosec} x + C$

6. Evaluate $\int \frac{\sec x + \cos x}{2 \cos x} dx$

Solution: We can write $\int \frac{\sec x + \cos x}{2 \cos x} dx = \frac{1}{2} \int (\sec^2 x + 1) dx = \frac{1}{2} (\tan x + x) + C$

7. Evaluate $\int \left[\frac{4}{x\sqrt{x^2-1}} + \frac{1+x+x^3}{1+x^2} \right] dx$

Solution: We can write

$$\begin{aligned} \int \left[\frac{4}{x\sqrt{x^2-1}} + \frac{1+x+x^3}{1+x^2} \right] dx &= 4 \sec^{-1} x + \int \frac{1}{1+x^2} dx + \int \frac{x(1+x^2)}{1+x^2} dx \\ &= 4 \sec^{-1} x + \tan^{-1} x + \frac{1}{2} x^2 + C \end{aligned}$$

Integration by Parts and Standard Integrals

1. Evaluate $\int \ln x dx$

Solution: We can write $\int \ln x dx = \ln x \int 1 dx - \int \frac{1}{x} x dx = x \ln x - x + c$

2. Evaluate $\int x^2 \tan^{-1} x dx$

Solution: We can write

$$\begin{aligned} \int x^2 \tan^{-1} x dx &= \tan^{-1} x \int x^2 dx - \int \frac{1}{1+x^2} \frac{x^3}{3} dx \\ &= \frac{x^3}{3} \tan^{-1} x - \frac{1}{3} \int \frac{x^3}{1+x^2} dx = \frac{x^3}{3} \tan^{-1} x - \frac{1}{3} \int \frac{x(x^2+1)-x}{1+x^2} dx + c \\ &= \frac{1}{3} x^3 \tan^{-1} x - \frac{1}{3} \times \frac{x^2}{2} + \frac{1}{3} \int \frac{x}{1+x^2} dx + c \\ &= \frac{1}{3} x^3 \tan^{-1} x - \frac{x^2}{6} + \frac{1}{6} \ln(1+x^2) + c \end{aligned}$$

3. $\int \frac{e^{m \tan^{-1} x}}{(1+x^2)^2} dx$

Solution. Let $\tan^{-1} x = z \Rightarrow \frac{1}{1+x^2} dx = dz$ and $x = \tan z$

So $\int \frac{e^{m \tan^{-1} x}}{(1+x^2)^2} dx = \int \frac{e^{mz}}{1+\tan^2 z} dz = \int e^{mz} \cos^2 z dz = \frac{1}{2} \int e^{mz} (1 + \cos 2z) dz$

$$\begin{aligned}
&= \frac{1}{2} \int e^{mz} dz + \frac{1}{2} \int e^{mz} \cos 2z dz \\
&= \frac{1}{2m} e^{mz} + \frac{1}{2} \times \frac{e^{mz}}{m^2 + 4} [m \cos 2z + 2 \sin 2z] + c \\
&= \frac{1}{2m} e^{m \tan^{-1} x} + \frac{e^{m \tan^{-1} x}}{2(m^2 + 4)} [m \cos(2 \tan^{-1} x) + 2 \sin(2 \tan^{-1} x)] + c
\end{aligned}$$

4. Evaluate $\int \sqrt{1+x^2} x^5 dx$

Solution: We can write $\int \sqrt{1+x^2} x^5 dx = \int \sqrt{1+x^2} x^4 \cdot x dx$

Let $1+x^2 = u$. Then $2x dx = du$ giving $x dx = (1/2) du$

$$\begin{aligned}
\text{Therefore, } \int \sqrt{1+x^2} x^5 dx &= \frac{1}{2} \int \sqrt{u} (u-1)^2 du = \frac{1}{2} \int \sqrt{u} (u^2 - 2u + 1) du \\
&= \frac{1}{2} \int (u^{5/2} - 2u^{3/2} + u^{1/2}) du
\end{aligned}$$

5. Evaluate $\int \frac{dx}{(1-x)\sqrt{1+x}}$

Solution: Substitute $1+x = u^2 \Rightarrow dx = 2u du$

$$\begin{aligned}
\text{So that we can write } \int \frac{dx}{(1-x)\sqrt{1+x}} &= \int \frac{2u}{(1-u^2+1)u} du = 2 \int \frac{1}{2-u^2} du \\
&= 2 \int \frac{1}{(\sqrt{2})^2 - (u)^2} du = 2(1/2\sqrt{2}) \log \frac{\sqrt{2}+u}{\sqrt{2}-u} + c = (1/\sqrt{2}) \log \frac{\sqrt{2}+\sqrt{1+x}}{\sqrt{2}-\sqrt{1+x}} + c
\end{aligned}$$

6. Evaluate $\int \frac{x^2+1}{x^4+x^2+1} dx$

$$\text{Solution: We can write } \int \frac{x^2+1}{x^4+x^2+1} dx = \int \frac{x^2 \left(1 + \frac{1}{x^2}\right)}{x^2 \left(x^2 + 1 + \frac{1}{x^2}\right)} dx = \int \frac{1 + \frac{1}{x^2}}{\left(x - \frac{1}{x}\right)^2 + 3} dx$$

$$\text{Let } x - \frac{1}{x} = z \Rightarrow \left(1 + \frac{1}{x^2}\right) dx = dz$$

Therefore, we get

$$\begin{aligned}
\int \frac{x^2+1}{x^4+x^2+1} dx &= \int \frac{dz}{z^2+3} = \frac{1}{\sqrt{3}} \tan^{-1} \frac{z}{\sqrt{3}} + c = \frac{1}{\sqrt{3}} \tan^{-1} \frac{x - \frac{1}{x}}{\sqrt{3}} + c \\
&= \frac{1}{\sqrt{3}} \tan^{-1} \frac{x^2-1}{\sqrt{3}x} + c
\end{aligned}$$

7. Evaluate $\int \frac{\sin^{-1} x}{(1-x^2)^{3/2}} dx$

$$\text{Solution: Let } \sin^{-1} x = z \Rightarrow \frac{1}{\sqrt{1-x^2}} dx = dz \text{ and } x = \sin z.$$

$$\begin{aligned}
\text{We can write } \int \frac{\sin^{-1} x}{(1-x^2)^{3/2}} dx &= \int \frac{\sin^{-1} x}{(1-x^2)^{1/2}(1-x^2)} dx = \int \frac{z}{1-\sin^2 z} dz \\
&= \int z \sec^2 z dz = z \tan z - \int \tan z dz + c = z \tan z + \ln(\cos z) + c \\
&= \sin^{-1} x \tan(\sin^{-1} x) + \ln \cos(\sin^{-1} x) + c \\
&= \frac{x}{\sqrt{1-x^2}} \sin^{-1} x + \ln \sqrt{1-x^2} + c
\end{aligned}$$

8. $\int \frac{1}{1+\sin x + \cos x} dx$

Solution: Let $I = \int \frac{1}{1+\sin x + \cos x} dx$.

Dividing both numerator and denominator by $\cos x$, we get $I = \int \frac{\sec x}{\sec x + \tan x + 1} dx$.

Let $\sec x + \tan x + 1 = z$. Then $(\sec x \tan x + \sec^2 x) dx = dz$ and so we get

$$\sec x(\sec x + \tan x) dx = dz \Rightarrow \sec x dx = \frac{dz}{z^2 - 1}$$

Therefore, we get

$$\begin{aligned}
I &= \int \frac{dz}{z(z-1)} = \int \left(\frac{1}{z-1} - \frac{1}{z} \right) dz = \ln|z-1| - \ln|z| + C \\
&= \ln \left| \frac{z-1}{z} \right| + C = \ln \left| \frac{\sec x + \tan x}{\sec x + \tan x + 1} \right| + C.
\end{aligned}$$

Alternative method

$$\text{We can write } 1 + \sin x + \cos x = 1 + \frac{2 \tan(x/2)}{1 + \tan^2(x/2)} + \frac{1 - \tan^2(x/2)}{1 + \tan^2(x/2)} = 2 + 2 \tan(x/2)$$

$$I = \int \frac{1}{1 + \sin x + \cos x} dx = \int \frac{1 + \tan^2(x/2)}{2 + 2 \tan(x/2)} dx = \frac{1}{2} \int \frac{\sec^2(x/2)}{1 + \tan(x/2)} dx$$

$$\text{Let } 1 + \tan \frac{x}{2} = z \Rightarrow \frac{1}{2} \sec^2 \frac{x}{2} dx = dz$$

$$\text{Therefore, } I = \int \frac{1}{z} dz = \ln|z| + C = \ln|1 + \tan(x/2)| + C$$

9. $\int \frac{1 + \sin x}{\sin x(1 + \cos x)} dx$

Solution: Given $\int \frac{1 + \sin x}{\sin x(1 + \cos x)} dx = \int \left(\frac{1}{\sin x(1 + \cos x)} + \frac{\sin x}{\sin x(1 + \cos x)} \right) dx$

$$= \int \left(\frac{1}{\sin x(1 + \cos x)} + \frac{1}{1 + \cos x} \right) dx = I_1 + I_2, \text{ say}$$

$$\text{Now } I_1 = \int \frac{1}{\sin x(1 + \cos x)} dx$$

$$\text{We can write } \sin x(1 + \cos x) = \frac{2 \tan \frac{x}{2}}{1 + \tan^2 \frac{x}{2}} \left(1 + \frac{1 - \tan^2 \frac{x}{2}}{1 + \tan^2 \frac{x}{2}} \right)$$

$$= \frac{2 \tan \frac{x}{2}}{1 + \tan^2 \frac{x}{2}} \left(\frac{1 + \tan^2 \frac{x}{2} + 1 - \tan^2 \frac{x}{2}}{1 + \tan^2 \frac{x}{2}} \right) = \frac{2 \tan \frac{x}{2}}{1 + \tan^2 \frac{x}{2}} \left(\frac{2}{1 + \tan^2 \frac{x}{2}} \right) = \frac{4 \tan \frac{x}{2}}{(1 + \tan^2 \frac{x}{2})(1 + \tan^2 \frac{x}{2})}$$

Let $\tan \frac{x}{2} = u \Rightarrow \frac{1}{2} \sec^2 \frac{x}{2} dx = du$

So
$$I_1 = \int \frac{1}{\sin x(1 + \cos x)} dx = \int \frac{(1 + \tan^2 \frac{x}{2}) \sec^2 \frac{x}{2}}{4 \tan \frac{x}{2}} dx$$

$$= 2 \int \frac{(1 + u^2)}{4u} du = \frac{1}{2} \int \frac{1}{u} du + \frac{1}{2} \int u du = \frac{1}{2} \ln|u| + \frac{u^2}{4} = \frac{1}{2} \ln \left| \tan \frac{x}{2} \right| + \frac{1}{4} \tan^2 \frac{x}{2}$$

$$I_2 = \int \frac{1}{1 + \cos x} dx = \frac{1}{2} \int \sec^2 \frac{x}{2} dx = \tan \frac{x}{2}$$

Hence, $I = I_1 + I_2$.

10. Evaluate $\int \frac{1}{\cos 3x - \cos x} dx$

Solution: Let $I = \int \frac{x^2}{(x \sin x + \cos x)^2} dx = \int x \sec x \frac{x \cos x}{(x \sin x + \cos x)^2} dx$

$$= x \sec x \int \frac{x \cos x}{(x \sin x + \cos x)^2} dx - \int \left[\frac{d}{dx} (x \sec x) \int \frac{x \cos x}{(x \sin x + \cos x)^2} dx \right] dx$$

$$= (x \sec x) I_1 - \int (\sec x + x \sec x \tan x) I_1 dx \quad \text{where } I_1 = \int \frac{x \cos x}{(x \sin x + \cos x)^2} dx$$

Let $x \sin x + \cos x = z \Rightarrow (x \cos x + \sin x - \sin x) dx = dz \Rightarrow x \cos x dx = dz$

Also, $\sec x + x \sec x \tan x = \frac{1}{\cos x} + x \frac{1}{\cos x} \frac{\sin x}{\cos x} = \frac{\cos x + x \sin x}{\cos^2 x}$

Therefore, we get $I_1 = \int \frac{1}{z^2} dz = -\frac{1}{z} = -\frac{1}{x \sin x + \cos x}$

Hence, $I = -\frac{x \sec x}{x \sin x + \cos x} + \int \sec^2 x dx = -\frac{x \sec x}{x \sin x + \cos x} + \tan x + C$

Alternative method

We can write $\int \frac{1}{\cos 3x - \cos x} dx = \int \frac{1}{2 \sin 2x \sin x} dx = \frac{1}{4} \int \frac{1}{\sin^2 x \cos x} dx$

$$= \frac{1}{4} \int \frac{\sin^2 x + \cos^2 x}{\sin^2 x \cos x} dx = \frac{1}{4} \int (\sec x dx + \csc x \cot x) dx$$

$$= \frac{1}{4} \ln |\sec x + \tan x| - \frac{1}{4} \csc x + C = \frac{1}{4} \ln \left| \tan \left(\frac{\pi}{4} + \frac{x}{2} \right) \right| - \frac{1}{4} \csc x + C$$